TRY THESE

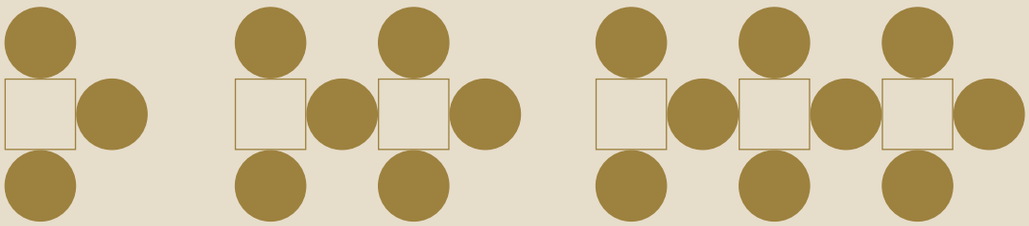
- Observe the following patterns and complete them.

  (i) 5, 8, 11, 14, ___ , ___ , ___

  (ii) If 15873 × 7 = 111111 and 15873 × 14 = 222222 then what is

  15873 × 21 =? and 15873 × 28 =?

- Draw the next two patterns and complete the table.



Pattern	1st	2nd	3rd	4th	5th
Squares	1	2	3		
Circles	3	6	9		

- Create your own pattern of shapes and prepare a table.

2.2.1 Patterns in Number Operations

In the first chapter on Whole Numbers, you have learnt about numbers that are multiplied by 1 or 0.

For example, we know, 57 × 1 = 57 and 43 × 0 = 0.

But we also know that this statement is true for all numbers (not just for the above two). So, can we say "any number" × 1 = "the same number" that we started with?

Algebra gives a way for writing such facts in a short and sweet way. We can write the above statement as $n \times 1 = n$, where n is a number. Here n on the left-hand side is just a letter that is used instead of saying "any number". The number on the right-hand side is the same n. This ensures that we get a correct statement!

In Algebra, we say that "n" is a *variable*. A variable is a symbol (usually an alphabet like n or x) that represents a number. Variables often help us to write briefly what we mean by a relation. In $n \times 1 = n$, left side number quantity 1 do not vary always. So we call it as *constant*.

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